

Dear Hiring Team,

I am excited to apply for the Summer Student Programme 2025 at CERN. As a third-year Theoretical Physics student at Trinity College Dublin in Ireland, I have a special passion for particle physics and am eager to contribute to one of the most remarkable scientific endeavours of our generation. Throughout my course, I have had a special interest in advanced classical mechanics and relativistic field theory and I plan to move to high energy physics in my 4th Year.

I believe I have a unique set of skills which would make me an invaluable addition to any research project: a passion for theoretical physics combined with skills in software development. I have spent time working on Python, ML models and have a working knowledge of Amazon Cloud Services which only enhance my skills covered in my computer simulation modules.

You will be able to see from my attached references that my programming skills have contributed greatly to the projects I was involved in. For example, my research internship last summer in Electronic Physics lab was the time where I have managed to successfully develop a Python-based application for spectroscopy measurements in addition to acquiring new skills in experimental physics and computational analysis using Origin and Python. On the first day of joining, I was up for a challenge and was given many ferromagnetic liquids to analyse. When I left the lab, the researchers could continue with using the new machine with the help of the application I wrote for them. Similarly, when I was asked to join the TechEd start-up my programming skills were invaluable to the successful launch. Please refer to the references attached to hear directly from the team members who I have worked closely with.

I would be deeply honoured to join CERN's mission, and I am committed to contributing my utmost to its pioneering work. It would be a privilege to be a part of and to learn from the best. I sincerely hope to bring my commitment and hard work to this exceptional opportunity.

And finally, I would like to add that my passion for Theoretical Physics started at a very early age. And I would like to dedicate my life's purpose not only to the research but also to making it more accessible to everyone. I feel this field of science is not promoted and explained enough for the young people. During the course of my career, I am hoping to find and create many new engaging ways to do just that.

Thank you for considering my application. You will find my contact details below.

Your Sincerely,

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